



checkCIF/PLATON report

You have not supplied any structure factors. As a result the full set of tests cannot be run.

THIS REPORT IS FOR GUIDANCE ONLY. IF USED AS PART OF A REVIEW PROCEDURE FOR PUBLICATION, IT SHOULD NOT REPLACE THE EXPERTISE OF AN EXPERIENCED CRYSTALLOGRAPHIC REFEREE.

No syntax errors found. CIF dictionary Interpreting this report

Datablock: 16369-ICSD

Bond precision: = 0.0000 Å

Wavelength=0.71073

Cell: a=5.53500 b=11.39910 c=5.15900

 alpha=90 beta=94.6000 gamma=90

Temperature: 0 K

	Calculated	Reported
Volume	324.454	324
Space group	I 2/a	I 1 2/a 1
Hall group	-I 2ya	?
Moiety formula	Ce Nb O4	?
Sum formula	Ce Nb O4	Ce1 Nb1 O4
Mr	297.03	0.00
Dx, g cm ⁻³	6.081	0.000
Z	4	4
Mu (mm ⁻¹)	17.210	0.000
F000	524.0	0.0
F000'	514.81	
h, k, lmax		
Nref		
Tmin, Tmax		
Tmin'		

Correction method= Not given

Data completeness=

Theta (max)=

R(reflections)=

wR2(reflections)=

S =

Npar=

The following ALERTS were generated. Each ALERT has the format

test-name_ALERT_alert-type_alert-level.

Click on the hyperlinks for more details of the test.

Alert level A

EXPT005_ALERT_1_A _exptl_crystal_description is missing
Crystal habit description.
The following tests will not be performed.
CRYSR_01

DIFF003_ALERT_1_A _diffrn_measurement_device_type is missing
Diffractometer make and type. Replaces _diffrn_measurement_type.

ATOM007_ALERT_1_A _atom_site_aniso_label is missing
Unique label identifying the atom site.

GEOM001_ALERT_1_A _geom_bond_atom_site_label_1 is missing
Label identifying the atom site 1.

GEOM002_ALERT_1_A _geom_bond_atom_site_label_2 is missing
Label identifying the atom site 2.

GEOM003_ALERT_1_A _geom_bond_distance is missing
Distance between atom sites 1 and 2.

GEOM006_ALERT_1_A _geom_angle_atom_site_label_2 is missing
Label identifying the atom site 2.

GEOM007_ALERT_1_A _geom_angle_atom_site_label_3 is missing
Label identifying the atom site 3.

GEOM008_ALERT_1_A _geom_angle is missing
Angle between atom sites 1, 2 and 3.

PLAT029_ALERT_3_A _diffrn_measured_fraction_theta_full value Low . 0.000 Why?

PLAT043_ALERT_1_A Calculated and Reported Mol. Weight Differ by .. 297.03 Check

PLAT183_ALERT_1_A Missing _cell_measurement_reflns_used Value Please Do !

PLAT184_ALERT_1_A Missing _cell_measurement_theta_min Value Please Do !

PLAT185_ALERT_1_A Missing _cell_measurement_theta_max Value Please Do !

PLAT197_ALERT_1_A Missing _cell_measurement_temperature Datum Please Add

PLAT198_ALERT_1_A Missing _diffrn_ambient_temperature Datum Please Add

PLAT699_ALERT_1_A Missing _exptl_crystal_description Value Please Do !

PLAT880_ALERT_1_A No datum for _diffrn_reflns_number supplied Please Do !

PLAT881_ALERT_1_A No Datum for _diffrn_reflns_av_R_equivalents ... Please Do !

Alert level C

PLAT041_ALERT_1_C Calc. and Reported SumFormula Strings Differ Please Check
Calc: Ce Nb O4
Rep.: Ce1 Nb1 O4

PLAT141_ALERT_4_C s.u. on a - Axis Small or Missing 0.00000 Ang.

PLAT142_ALERT_4_C s.u. on b - Axis Small or Missing 0.00000 Ang.

PLAT143_ALERT_4_C s.u. on c - Axis Small or Missing 0.00000 Ang.

PLAT145_ALERT_4_C s.u. on beta Small or Missing 0.0000 Degree

PLAT151_ALERT_1_C No s.u. (esd) Given on Volume Please Do !
Ce1 Nb1 O1 O2

● Alert level G

FORMU01_ALERT_2_G There is a discrepancy between the atom counts in the
_chemical_formula_sum and the formula from the _atom_site* data.
Atom count from _chemical_formula_sum: Ce1 Nb1 O4
Atom count from the _atom_site data: Ce1 Nb1

CELLZ01_ALERT_1_G Difference between formula and atom_site contents detected.

CELLZ01_ALERT_1_G ALERT: Large difference may be due to a
symmetry error - see SYMMG tests
From the CIF: _cell_formula_units_Z 4
From the CIF: _chemical_formula_sum Ce1 Nb1 O4
TEST: Compare cell contents of formula and atom_site data
WARNING: Unexpected atom type is in site list: Ce3+
WARNING: Unexpected atom type is in site list: Nb5+
WARNING: Unexpected atom type is in site list: O2-
Note that site occupancies missing: assumed 1.0
WARNING: Formula and atom_type_symbol element names mismatch.

atom	Z*formula	cif sites	diff
Ce	4.00	0.00	4.00
Nb	4.00	0.00	4.00
O	16.00	0.00	16.00

WARNING: Site labels do not match formula elements

PLAT004_ALERT_5_G	Polymeric Structure Found with Maximum Dimension	3	Info
PLAT005_ALERT_5_G	No Embedded Refinement Details Found in the CIF	Please	Do !
PLAT128_ALERT_4_G	Alternate Setting for Input Space-group I2/a	I2/c	Note
PLAT720_ALERT_4_G	Number of Unusual/Non-Standard Labels	1	Note
	O1		
PLAT794_ALERT_5_G	Tentative Bond Valency for Ce1 (III)	3.34	Info
PLAT794_ALERT_5_G	Tentative Bond Valency for Nb1 (V)	4.76	Info
PLAT808_ALERT_5_G	No Parseable SHELXL Style Weighting Scheme Found	Please	Check
PLAT882_ALERT_1_G	No Datum for _diffn_reflns_av_unetI/netI	Please	Do !
PLAT883_ALERT_1_G	Absent Datum for _atom_sites_solution_primary ..	Please	Do !
PLAT980_ALERT_1_G	No Anomalous Scattering Factors Found in CIF ...	Please	Check

19 **ALERT level A** = Most likely a serious problem - resolve or explain
0 **ALERT level B** = A potentially serious problem, consider carefully
6 **ALERT level C** = Check. Ensure it is not caused by an omission or oversight
13 **ALERT level G** = General information/check it is not something unexpected

25 ALERT type 1 CIF construction/syntax error, inconsistent or missing data
1 ALERT type 2 Indicator that the structure model may be wrong or deficient
1 ALERT type 3 Indicator that the structure quality may be low
6 ALERT type 4 Improvement, methodology, query or suggestion
5 ALERT type 5 Informative message, check

It is advisable to attempt to resolve as many as possible of the alerts in all categories. Often the minor alerts point to easily fixed oversights, errors and omissions in your CIF or refinement strategy, so attention to these fine details can be worthwhile. It is up to the individual to critically assess their own results and, if necessary, seek expert advice.

duplicate check

A reduced cell check using CCDC's cellCheckCSD service has found that one or more structures in this CIF are similar to those previously published in the CSD or the ICSD.

DATABLOCK: 16369-ICSD

- ICSD number: 117798



[Cell Parameters for 117798: 7.2609,11.4032,5.1621(90,130.5301,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 16369



[Cell Parameters for 16369: 5.535,11.3991,5.159(90,94.6,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235733



[Cell Parameters for 235733: 7.2448,11.3685,5.1515(90,130.543,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235734



[Cell Parameters for 235734: 7.318,11.4839,5.2036(90,130.525,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235735



[Cell Parameters for 235735: 7.2526,11.3703,5.1532(90,130.559,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235736



[Cell Parameters for 235736: 7.2729,11.389,5.1617(90,130.706,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235737



[Cell Parameters for 235737: 7.249,11.3633,5.1536(90,130.572,90)]

DATABLOCK: 16369-ICSD

- ICSD number: 235738



[Cell Parameters for 235738: 7.274,11.389,5.1622(90,130.733,90)]

